

1. A new data point lies exactly between two classes with equal support. How should it be classified?
2. A classification method assigns labels based on nearby data points. What is the core idea of this approach?
3. A classification algorithm requires selecting a parameter k . What does this parameter control?
4. A model uses $k = 1$ to classify a new observation. How is the class determined?
5. A model considers three nearest neighbors with labels $[0, 1, 0]$. What class will be predicted?
6. Why can choosing an even value of k (e.g., $k = 2$) be problematic in binary classification?
7. In a nearest-neighbor classification method, what determines how many data points influence the decision?
8. What is the general effect of increasing k in a nearest-neighbor model?
9. What is the general effect of decreasing k in a nearest-neighbor model?
10. A machine learning model requires input in the shape (number of samples, number of features). How should a single data point with one feature be formatted?
11. A plot shows predicted class values across a range of inputs. What does such a plot represent?